AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2014

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. Complete the given table with the percentage of oxygen and carbon dioxide in the exhaled air. (2 Marks)

	Inhaled Air	**Exhaled Air**
Oxygen	21%	
Carbon dioxide	0.03%	

b. Give any TWO examples of involuntary action related to our body. (2 Marks)

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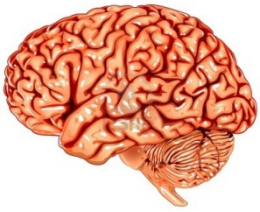
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Q.2. (Total 4 Marks)

The flow chart below represents osmoregulation in man. Fill in the given boxes with appropriate information.



stimulates

What happens to osmotic pressure?

_____

Loss of water through sweating

Name of the hormone released:

_____

What does the above hormone do?

_____

What effect does the above hormone have on the amount of urine?

_____

Q.3. (Total 4 Marks)

a. The presence of discs between individual vertebral bones is a unique feature of the backbone. Why is the backbone, in this aspect, different from other bones of the body? (2 Marks)

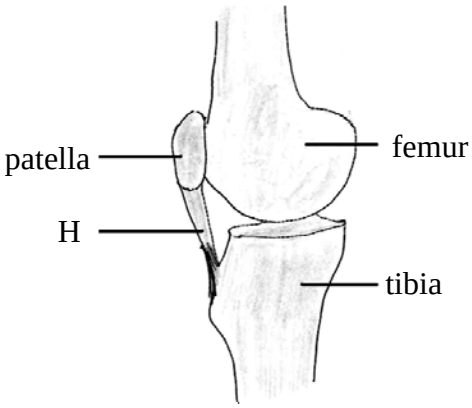
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b. The diagram shows part of the skeletal system. (2 Marks)



Identify structure H and state its importance in the human body.

_____

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Q.4. (Total 4 Marks)

a. State any TWO possible modes of transfer of AIDS. (2 Marks)

_____

_____

_____

_____

b. Briefly describe the process through which nitrogen is returned to the atmosphere. (Hint: Refer Nitrogen Cycle) (2 Marks)

_____

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Q.5. (Total 4 Marks)

According to a theory of evolution, “The variants better adapted to the environment are successful.”

i. Name the scientist who proposed this theory.

ii. Give THREE features of the animal shown below which make it adaptable to the polar environment.



Q.6. (Total 5 Marks)

a. Describe the importance of fermentation products in our daily life. (3 Marks)

b. State the contribution of the following scientists in the field of pharmacology. (2 Marks)

i. Alexander Fleming
ii. Joseph Lister

Q.7. (Total 7 Marks)

EITHER

a. What do you understand by the term, ‘negative feedback’? Explain how adrenaline helps us to prepare in an emergency situation.

OR

b. Explain how our skin is able to maintain our body temperature with particular reference to the structural features of skin.

Q.8. (Total 8 Marks)

EITHER

a. Describe the process of sperm and egg cell production in a male and a female rabbit respectively.

OR

b. Describe any FOUR causes of water pollution in our ecosystem.

END OF PAPER

Please use this page for rough work